Bayesian Machine Learning

An Introduction to the Unit

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2026-09-29

Slide 1 In this Episode ...

- **Welcome to "Bayesian Machine Learning" — CM32024 & CM52037**
- Logistics and unit overview
- Plan of lectures & labs (Semester 1)
- Assessment
- Introduction to Bayesian statistics
  - The "What?" and the "Why?"
  - Historical development
  - Philosophy
  - ... and, finally, Bayes' rule!
- Suggested background reading
- Looking ahead

Course Overview

Slide 2 Lectures and Labs (Semester 1)

- **Lectures**
  - **Lecture 'A'** (1 hour): Tuesday, 12.15am, 8W 2.5
  - **Lecture 'B'** (1 hour): Thursday, 11.15pm, 1WN 3.11
  - 18 lectures of content overall, in weeks 1–10
  - No lectures or labs planned in focus week (week 6) (Nov 2–6) or week 11 (Dec 7–11)
- **Labs**
  - 2 hours per week, Wednesday, 10:15am, 3E 3.1
  - 9 labs, 1 drop-in session (no lab this week or in focus week. Last week is a drop-in session for catching up)
  - Labs consist of a mixture of written exercises and Jupyter notebook implementation exercises.
  - Assistance: Lecturer (me), and (hopefully) a teaching assistant (TA) will be present in the lab each week.

Slide 3 Semester 1 Timetable (Details Subject to Change)

Week No.	W/C	No.	Topic	Date	Time	Room
1	Sep 28	Lecture 1	Bayesian Machine Learning	Tue Sep 29	12:15pm	8W 2.5
		No Lab	—	—	—	—
		Lecture 2	Probabilistic Thinking	Thu Oct 01	11:15am	1WN 3.11
2	Oct 05	Lecture 3	Bayesian Statistical Inference	Tue Oct 06	12:15pm	8W 2.5
		Labsheet 1	Bayesian Bandit Exercise	Wed Oct 07	10:15am	3E 3.1
		Lecture 4	Adventures in Inference	Thu Oct 08	11:15am	1WN 3.11
3	Oct 12	Lecture 5	Bayesian Linear Regression	Tue Oct 13	12:15pm	8W 2.5
		Labsheet 2	Regression Exercise	Wed Oct 14	10:15am	3E 3.1
		Lecture 6	Marginalisation & Occam's Razor I	Thu Oct 15	11:15am	1WN 3.11
4	Oct 19	Lecture 7	Marginalisation & Occam's Razor II	Tue Oct 20	12:15pm	8W 2.5
		Labsheet 3	Marginal Likelihood Exercise	Wed Oct 21	10:15am	3E 3.1
		Lecture 8	Markov Chain Monte Carlo (MCMC)	Thu Oct 22	11:15am	1WN 3.11
5	Oct 26	Lecture 9	Hamiltonian Monte Carlo (HMC)	Tue Oct 27	12:15pm	8W 2.5
		Labsheet 4	Prediction Exercise	Wed Oct 28	10:15am	3E 3.1
		Lecture 10	Probabilistic Programming	Thu Oct 29	11:15am	1WN 3.11
6	Nov 02	*Focus Week*	—	—	—	—
7	Nov 09	Lecture 11	Hierarchical Models I	Tue Nov 10	12:15pm	8W 2.5
		Labsheet 5	MCMC Exercise	Wed Nov 11	10:15am	3E 3.1
		Lecture 12	Hierarchical Models II	Thu Nov 12	11:15am	1WN 3.11
8	Nov 16	Lecture 13	Automatic Relevance Determination (ARD)	Tue Nov 17	12:15pm	8W 2.5
		Labsheet 6	Hierarchical Models Exercise	Wed Nov 18	10:15am	3E 3.1
		Lecture 14	Sparse Bayesian Learning and Inverse Models	Thu Nov 19	11:15am	1WN 3.11
9	Nov 23	Lecture 15	Bayesian Models for Time Series (SMC)	Tue Nov 24	12:15pm	8W 2.5
		Labsheet 7	Sparsity Exercise	Wed Nov 25	10:15am	3E 3.1
		Lecture 16	Bayesian Anomaly Detection	Thu Nov 26	11:15am	1WN 3.11
10	Nov 30	Lecture 17	Model Criticism and Calibration	Tue Dec 01	12:15pm	8W 2.5
		Labsheet 8	Time Series Exercise	Wed Dec 02	10:15am	3E 3.1
		Lecture 18	Seven Pillars of Bayesian Wisdom	Thu Dec 03	11:15am	1WN 3.11
11	Dec 07	*No Lecture*	—	—	—	—
		Drop-in	Drop-in Session	Wed Dec 09	10:15am	3E 3.1
		No Lecture	—	—	—	—

Slide 4 Assessment

- **100% In-Person Exam (Summer)**
  - All content from lectures and labs is examinable (unless stated otherwise)
  - The exam will be held in the summer exam period, in-person.
  - Mixture of theory and practical questions.

- Lectures, labs, and practice questions will prepare you for the exam.

- Some differentiation between CM32024 and CM52037

Slide 5 **Course Materials**

- **Media**
  - PDF of lecture slides available in advance (at least 24h before)
  - Accessibile HTML version of slides (e.g. for screen readers)
  - Slides source code (LaTeX)
  - Occasional code from demos
  - Lecture videos made available within 24 hours (ideally)
  - No separate lectures notes
- **Further help**
  - Ask at the Wednesday labs
  - Ask your peers
  - Post in the Moodle general discussion forum
  - As a last resort, email me!

Motivation: Why “Bayesian”?

Slide 6 **Motivation: Bayesian Statistics — What, and Why?**

What is it?

- “Bayesian” implies that the rules of probability may be applied to model all sources of uncertainty
- By contrast, in orthodox or frequentist statistics, only intrinsically random variables are treated probabilistically

Why is this distinction important?

- This is more than a philosophical distinction!
- The Bayesian viewpoint is consistent with the nature of real-world uncertainty
- It also offers a range of advantageous features when “learning” from data
- But there is a drawback: many of the necessary calculations are problematic to undertake and must be approximated

A Brief History

Slide 7 A Brief History



quodque ſolum, certa nitri ſigna præbere, ſed plura concurrere debere, ut de vero nitro producto dubium non relinquatur.

LII. *An Eſſay towards ſolving a Problem in the Doctrine of Chances. By the late Rev. Mr.* Bayes, *F. R. S. communicated by Mr.* Price, *in a Letter to* John Canton, *A. M. F. R. S.*

Dear Sir,

Read Dec. 23, 1763.

I Now ſend you an eſſay which I have found among the papers of our deceaſed friend Mr. Bayes, and which, in my opinion, has great merit, and well deſerves to be preſerved. Experimental philoſophy, you will find, is nearly intereſted in the ſubject of it; and on this account there ſeems to be particular reaſon for thinking that a communication of it to the Royal Society cannot be improper.

He had, you know, the honour of being a member of that illuſtrious Society, and was much eſteemed by many in it as a very able mathematician. In an introduction which he has writ to this Eſſay, he ſays, that his deſign at firſt in thinking on the ſubject of it was, to find out a method by which we might judge concerning the probability that an event has to happen, in given circumſtances, upon ſuppoſition that we know nothing concerning it but that, under the ſame circum-

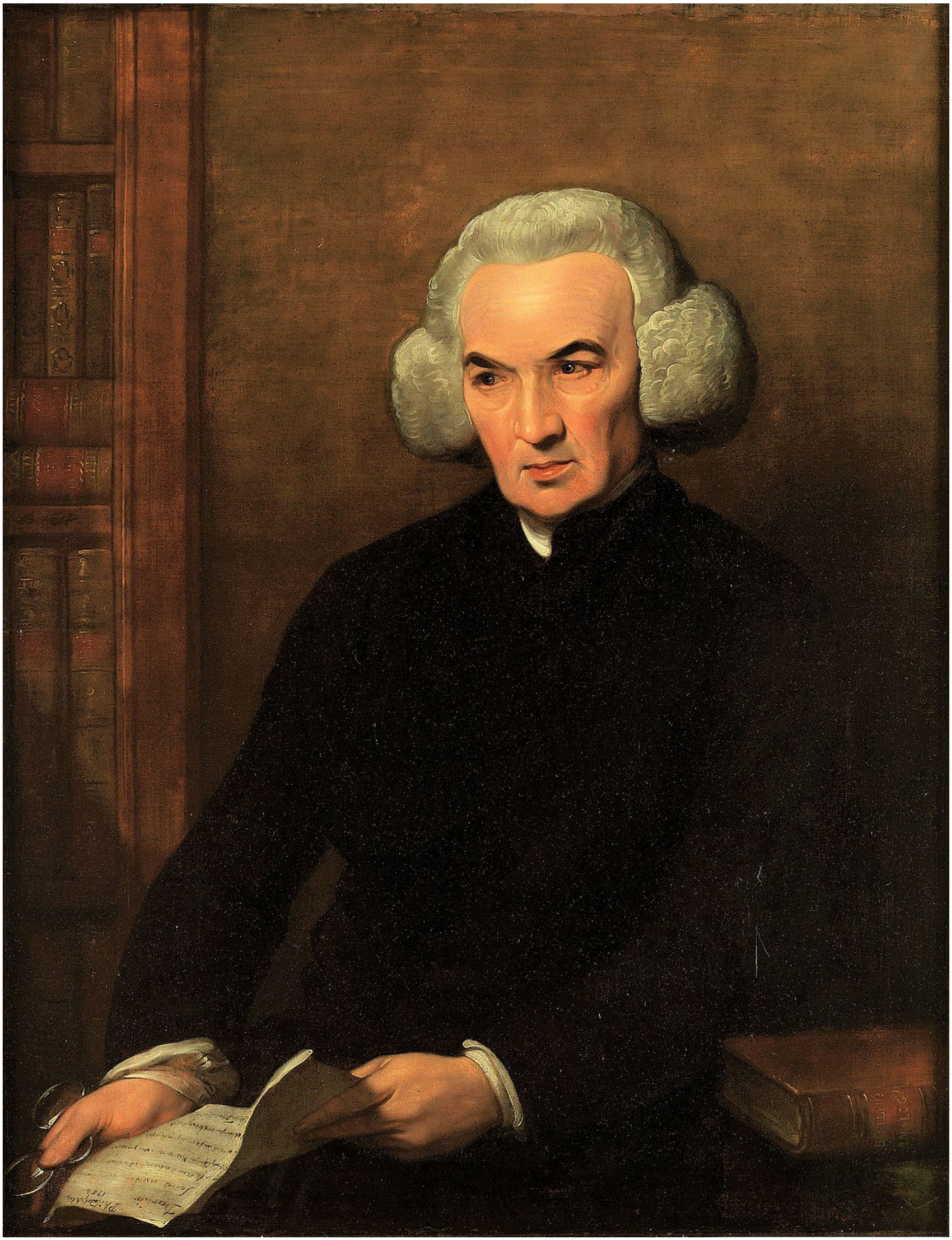






Photo by Jack Engeman

Richard T. Cox

is Professor of Physics and was formerly Dean of Arts and Sciences at The Johns Hopkins University. He was Professor of Physics at New York University before coming to Johns Hopkins. He is the author of *Time, Space and Atoms* and *Statistical Mechanics of Irreversible Change* and papers in journals of physics and biology.

"[This book] is, in my opinion one of the most important ever written on the foundations of probability theory, and the greatest advance in the conceptual, as opposed to the purely mathematical, formulation of the theory since Laplace."

—E. T. Jaynes, American Journal of Physics

The Johns Hopkins Press Baltimore 18, Maryland

ISBN 0-8018-6982-X





Slide 8 A Brief History of Probability: Jacob Bernoulli



- Jacob Bernoulli (1655-1705)
- In *Ars Conjectandi* (1713), he developed a theory of the relationship between the number of occurrences n of a binary event with probability p across N trials

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Slide 9 Practical Illustration

At this point, a bowl of red and blue balls is typically introduced …



Slide 10 A Brief History of Probability: Jacob Bernoulli

- Jacob Bernoulli (1655-1705)
- In *Ars Conjectandi* (1713), he developed a theory of the relationship between the number of occurrences n of a binary event with probability p across N trials



- He also recognised that the inverse of this problem was of interest: *i.e.* when n and N are known, what is p?
- This is the problem we now call statistical inference

Slide 11 Inverse Probability

- I have a bowl of $N = 6$ balls, an unknown mix of red and blue
- You draw a red one
- What is the proportion of red balls in the bowl?

In the bag of 6	Draw	Probability p
● ● ● ● ● ●	●	$1/6$
● ● ● ● ● ●	●	$1/3$
● ● ● ● ● ●	●	$1/2$
● ● ● ● ● ●	●	$2/3$
● ● ● ● ● ●	●	$5/6$
● ● ● ● ● ●	●	1

Slide 12 A Brief History of Probability: The Reverend Thomas Bayes

- Thomas Bayes (1701–1761) developed the first workable theory of "inverse probability" (in rather non-general form)
- It was edited and published posthumously (1763) by his literary executor Richard Price (1723–1791)

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PROBLEM.

Given the number of times in which an unknown event has happened and failed: *Required* the chance that the probability of its happening in a ſingle trial lies ſomewhere between any two degrees of probability that can be named.

Slide 13 A Brief History of Probability: Pierre-Simon Laplace



- Pierre-Simon Laplace (1749–1827)
- In *Essai Philosophique sur les Probabilités* (1814), Pierre-Simon Laplace formalised (independently of Bayes) a mathematical, probabilistic, system of inductive reasoning
- It is recognisably “Bayesian”
- He famously said: “The theory of probability is simply common sense reduced to calculus.”
- He was also Napoleon’s examiner at the École Militaire, and later Minister of the Interior of France (though he only lasted 6 weeks!)

Slide 14 Probability According to Laplace

In Laplace's system (simplified), if A and B are binary propositions (either true or false), and $\bar{A}$ denotes "not A" ($\bar{B}$ is "not B"), then just two axioms are required:

- The sum rule:

$$P(A) = P(A,B) + P\left(A,\bar{B}\right)$$

- The product rule:

$$P(A,B) = P\left(A|B\right) P(B) = P\left(B|A\right) P(A)$$

From these, we can derive Bayes' rule:

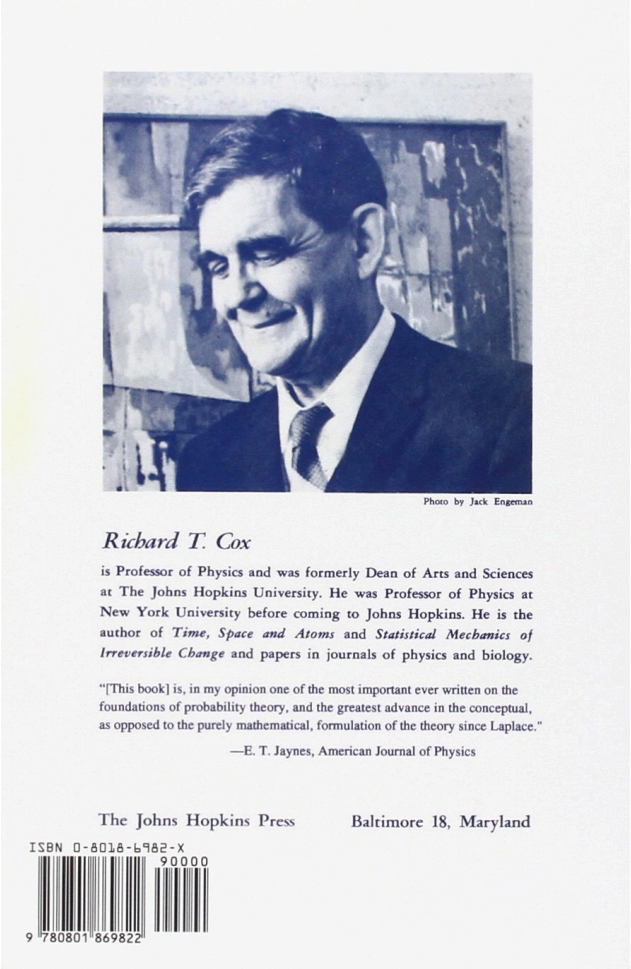
$$P\left(B|A\right) = \frac{P\left(A|B\right) P(B)}{P(A)} P\left(\text{Unknowns}|\text{Data}\right) = \frac{P\left(\text{Data}|\text{Unknowns}\right) P(\text{Unknowns})}{P(\text{Data})}$$

Slide 15 Orthodox, or "Frequentist", Statistics

- To Laplace *et al*, the probability $P(B)$ represented the degree of belief in, or plausibility of, the proposition (or hypothesis) B
- This interpretation was soon rejected as being too subjective
- Instead, probability came to be defined as the long run relative frequency of B occurring in an infinite set of trials
  - the implication is that B must **intrinsically** be a random variable
  - this is the orthodox or frequentist viewpoint
- The Bayesian view of probability (as plausibility) fell out of favour and, in the early 1900s, was rejected entirely ...

Slide 16 A Brief History of Probability: Richard T. Cox

- Richard T. Cox (1898–1991)



- *Probability, Frequency and Reasonable Expectation* (1946): Back to first principles
- Assumption: degrees of belief or plausibility can be represented by (any) real numbers
- Based on minimal desiderata, he sought a logically consistent system of rules for manipulating plausibilities
- Demonstrated that there was a single, unique, such system — and this was isomorphic to probability theory

The Modern Bayesian Viewpoint

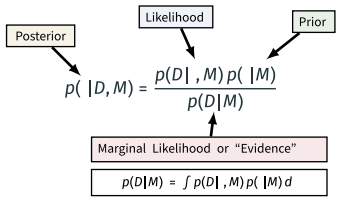
Slide 17 Conclusion: Probability Can Represent Plausibility

- Cox showed that belief functions were entirely consistent with probability theory, and its associated tool-set:
  - the sum and product rules
  - Bayes' theorem
- The Bayesian viewpoint is consequently that probability may be safely applied to model all forms of uncertainty, not just explicitly "random" variables
- For example, uncertainty such as:
  - the outcome of a flip of a coin

- whether a given patient has COVID-19
- whether it will rain during the next hour
- the value of a measured physical constant (*e.g.* the mass of the moon)
- the values a model parameter might take
- the choice of model itself

Slide 18 The Bayesian View in One Slide

If $\mathcal{D}$ is our observed data set, and $\boldsymbol{\theta}$ encapsulates all the unknowns associated with a model $\mathcal{M}$ intended to describe that data, then Bayesian Inference is:



Suggested Reading

Slide 19 Reading: Bayesian Statistics & Machine Learning

- (Bishop 2006) https://www.microsoft.com/en-us/research/people/cmbishop/#!prml-book
- (MacKay 2003) http://www.inference.org.uk/itprnn/book.html
- (Barber 2012) http://www.cs.ucl.ac.uk/staff/d.barber/brml/
- (Murphy 2022) http://probml.github.io/book1

Slide 20 Reading: Bayesian Probability and Statistics

- (Jaynes 2003)
- (Sivia and Skilling 2006)
- (Gelman et al. 2014)
- **“** Statistics — it’s just uncertainty laundering!**”** (Andrew Gelman). Watch his lecture “**Crimes against data**” at the University of Bath in 2016: https://www.youtube.com/watch?v=fc1hkFC2c1E
- (McElreath 2023). Note: this book is unfortunately focused on “R”, but has much good content.

Slide 21 Reading: Papers (Mainly of Historic Interest)

(Cox 1946) [Formulation of Bayesian statistics: available at Google Scholar]

(Loredo 1990) [Excellent review: accessible at Cornell]

(Bretthorst 1990) [Excellent introductory review: Google Scholar]

(Lindley 2000) [Interesting philosophical overview and discussion: Google Scholar]

(Gelman 2008)

These recommended examples cover Bayesian inference in general. References to other introductory papers relevant to specific Bayesian sub-topics will appear in future lectures and notes.

Slide 22 Looking Ahead (Semester 1) ...

- Bayesian inference
- Linear regression
- Model selection

- Markov Chain Monte Carlo (MCMC)
- Probabilistic programming (**NumPyro**, PyMC, Stan)
- Hierarchical models
- Sparse models
- Time series
- Calibration and model evaluation

Slide 23 Looking Ahead (Semester 2, With Some Uncertainty) ...

- Deterministic approximation (variational methods)
- The Gaussian Process (GP)
- Bayesian neural networks and deep learning
- Amortised Inference and Simulation-based Inference
- Bayesian optimisation
- Variational Autoencoders (VAE)
- Normalising flows

Barber, David. 2012. *Bayesian Reasoning and Machine Learning*. Cambridge University Press.

Bishop, Christopher M. 2006. *Pattern Recognition and Machine Learning*. Springer.

Bretthorst, G Larry. 1990. “An Introduction to Parameter Estimation Using Bayesian Probability Theory.” In *Maximum Entropy and Bayesian Methods*, 53–79. Springer.

Cox, Richard T. 1946. “Probability, Frequency and Reasonable Expectation.” *American Journal of Physics* 14 (1): 1–13.

Gelman, Andrew. 2008. “Objections to Bayesian Statistics.” *Bayesian Analysis* 3 (3): 445–49.

Gelman, Andrew, John B Carlin, Hal S Stern, and Donald B Rubin. 2014. *Bayesian Data Analysis (Third Edition)*. CRC Press.

Jaynes, Edward T. 2003. *Probability Theory: The Logic of Science*. Cambridge University Press.

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MacKay, David J. C. 2003. *Information Theory, Inference and Learning Algorithms*. Cambridge University Press.

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Murphy, Kevin P. 2022. *Probabilistic Machine Learning: An Introduction*. MIT Press.

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